MIL-D-81549A(AS)
22 February 1971
Superseding
MIL-D-81549(AS)
26 July 1967

MILITARY SPECIFICATION

DETONATOR, ELECTRIC, MARK 83 MOD 0

1. SCOPE

1.1 This document governs the manufacture, assembly and preparation for delivery of the MARK 83 MOD 0 electric detonator and the methods of inspection and tests upon which product acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this document to the extent specified herein.

SPECIFICATIONS

Military

MIL-D-21625	Design and Evaluation of Cartridges for Cartridge Actuated Devices
MIL-L-3055	Lead Azide
MIL-P-387	Pentaerythrite Tetranitrate (PETN)
MIL-P-116	Preservation, Methods of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

FSC 1377

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MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for

DRAWINGS AND ASSOCIATED LISTS

Naval Air Systems Command	(Code Ident No. 30003)
LD 537917	Detonator, Electric, MARK 83 MOD 0, List of Drawings and Specifications (Formerly Code Ident No. 10001)
LD 537918	Container, Inner, for Detonator, Electric, MARK 83 Type, List of Drawings and Specifications (Formerly Code Ident No. 10001)
LD 537921	Container, Outer, for Detonator, Electric, MARK 83 Type, List of Drawings and Specifications (Formerly Code Ident No. 10001)
1558439	Ignition Charge (Formerly Code Ident No. 10001)
2299542	Ignition Assembly (Formerly Code Ident No. 10001)
2299543	Plug and Bridgewire Assembly (Formerly Code Ident No. 10001)
2406376	Detonator, Electric, MARK 83 MOD 0, Assembly (Formerly Code Ident No. 10001)
2406508	Flash Charge (Formerly Code Ident No. 10001)

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PUBLICATIONS

BUSANDA Instruction 4030.17	Preservation, Packaging, Packing and Marking; Navy Policy for
Bureau of Naval Weapons	
OP 5	Ammunition Ashore (This publication is not furnished to bidders and contractors by the procuring activity. Contractors should contact the Government Inspector for use of the publication.)
WR-43	Preparation of Quality Assurance Provisions

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-78	Transportation, Interstate Commerce Commission, Explosives and Other Dangerous Articles

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publication should request the latest edition and supplements thereto.)

3. REQUIREMENTS

3.1 General.

3.1.1 The MARK 83 MOD 0 electric detonator covered by this specification shall be manufactured in accordance with LD 537917 and all documents listed thereon.

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3.2 Material. All materials used in the manufacture of the MARK 83 MOD 0 electric detonator shall conform strictly with the specifications referenced on the respective Naval Air Systems Command drawings unless specific approval in writing covering a departure therefrom has been obtained from the Navy cognizant design activity for cartridges prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.2.1 Preproduction sample. Unless otherwise specified by the procuring activity, a preproduction sample of 76 MARK 83 MOD 0 electric detonators shall be manufactured using the same methods and procedures proposed for the production lot. The sample shall be tested as specified in 4.3.2 and is for the purpose of determining that the contractor, prior to starting production, is capable of producing items which comply with the technical requirements of the contract.

3.3 Primary components. For the purposes of this specification, the PETN, MIL-P-387, is defined as a primary component. See 4.3.3.

3.4 Insulation resistance. Prior to the assembly of the bridgewire across the ground ends of the lead wires, the insulation resistance between the lead wires shall be a minimum of 50 megohms at a relative humidity of 80 percent or less. See 4.7.1.

3.5 Bridgewire. The bridgewire shall be soldered to the ground ends of the lead wires. The solder and soldering process shall be as specified on Drawing 2299543.

3.6 Bridgewire assembly, continuity. The bridgewire shall glow when subjected to the continuity test of 4.7.2.

3.7 Flash charge. The flash charge shall be compounded and loaded in accordance with Drawings 2406508 and 2299542, respectively. The charge shall completely cover the bridgewire.

3.8 Ignition charge. The ignition charge shall be compounded by a method which maintains the weight composition of nonvolatile constituents as specified on Drawing 1558439. The finished composition when ready for buttering into the detonator collar shall be a uniformly wetted slurry in which all particles are wetted. The consistency shall be such that the liquid will not separate from the mixture when the dish is tilted.

3.8.1 Loading of ignition charge. The technique of loading the ignition charge shall be such as to avoid damage to the bridgewire. The charge shall be smoothed off flush with the top of the collar and all of

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the charge material shall be removed from the exterior lateral areas of the collar. Allow to dry as specified on Drawing 2299542 before assembly with other components of the detonator. The dried charge shall be free of cracks, voids, and separation from the collar.

3.9 Booster and main charges.

3.9.1 Acceptance and loading. The lead azide used in the booster charge and the PETN used in the main charge shall be accepted in accordance with MIL-L-3055 and MIL-P-387 Class 4, respectively. The charges shall be loaded in accordance with the loading procedures specified on Drawing 2406376.

3.9.2 Quantity. The booster and the main charge quantities shall be as specified on Drawing 2406376. The specified charge quantities shall be determined by weight rather than volume measurements.

3.10 Detonator inspection.

3.10.1 Visual inspection. The complete detonator shall be free of the following visible defects: dents, holes, deep scratches, split or cracked lip in the crimped area, sharp edges; and all other defects which could conceivably prevent entry into the device of intended application, or affect performance or resistance to moisture. See 4.7.4.

3.10.2 Bridge circuit resistance. The resistance of the bridge circuit in completely assembled detonators, when measured electrically as specified in 4.7.3, shall be not less than 0.057 ohm nor more than 0.097 ohm.

3.10.3 X-ray. Detonators shall be free of imperfections in assembly when examined by X-ray in accordance with 4.7.5.

3.11 Ballistic requirements. Detonators from each production lot, selected in accordance with 4.7.6.1, shall meet the requirements listed below when subjected to the test of 4.7.6.1.

3.11.1 Output. The minimum depth (lower(L)limit) of indentation produced in the steel block shall be 0.033 inch. See 4.7.6.2.1.

3.11.2 Misfire. There shall be no misfires. See 4.7.6.2.2.

3.11.3 Hangfire. There shall be no hangfires. See 4.7.6.2.3.

3.12 Workmanship. Detonators shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention should be directed to dimensions, finishes and coatings.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. Unless otherwise specified, and when applicable, the sampling plans and procedures used in the determination of the acceptability of products submitted by a supplier shall be in accordance with the provisions of MIL-STD-105 or MIL-STD-414, whichever applies.

4.3 Lots.

4.3.1 Inspection lots. Inspection lot definition, formation, and size shall be in accordance with MIL-STD-105 or MIL-STD-414, whichever applies.

4.3.2 Preproduction sample. Before entering into quantity production, an acceptable preproduction sample of 76 MARK 83 MOD 0 electric detonators shall be prepared. Of these detonators, 66 shall be expended in the tests listed below and 10 shall be retained for investigative purposes. A preproduction sample acceptable for environmental and functional testing shall be defined as a sample which has met the requirements of paragraphs 3.1 through 3.10.3 of this specification. The contractor shall deliver the detonators to the activity designated by the contracting agency for the test. Preproduction sample acceptance shall consist of the following tests as prescribed in Table 3 of MIL-D-21625: 15 "g" shock (6), vibration (6), and temperature and humidity cycling (THC)(18) (except that THC shall be for 14 days in lieu of 28 days). The number in parentheses denotes the number of detonators allocated for test. These detonators, after exposure to environmental treatments, together with 12 untreated detonators shall be conditioned at 70°F ± 5°F for a minimum of six hours and fired in accordance with paragraph 4.7.6.1. In addition, 24 detonators shall be temperature conditioned for a minimum of six hours (12 at -65°F ± 5°F and 12 at +160°F ± 5°F) and fired as prescribed in 4.7.6.1. All detonators shall produce results within the limit set forth in 3.11.1 and shall meet the requirements of 3.11.2 and 3.11.3. The failure of any detonator to meet the requirements stated above shall cause rejection of the preproduction sample. Any quantity production prior to notification by the contracting agency of preproduction lot acceptability shall be at the contractor's risk. The preproduction detonators will not be applied as a part of the quantity specified for delivery by the contract.

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4.3.3 Production lot. Unless otherwise specified in the contract, the minimum and maximum production lot size shall be 501 and 800 detonators, respectively. Detonators required for all test purposes are selected from the production lot and will not be applied as a part of the quantity specified for delivery by the contract. Only primary components from a single lot shall be used in a production lot of detonators; however, one primary component production lot may be used in more than one detonator production lot. Detonators selected for the ballistic test shall be delivered to the activity designated by the contracting agency.

4.4 Gages. The contractor shall provide gages as necessary to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings.

4.5 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as Critical, Major or Minor in accordance with WR-43. Tests and examinations that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of test or examination. Tests and examinations which are not annotated with a classification code are classified minor.

4.6 Quality conformance inspection. Quality conformance inspection shall consist of verification of the characteristics classified on the drawings of LD 537917 and the following requirements and tests:

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Item	Nature of Requirement Verification	Applicable Paragraphs	Classification
Primary Component	Single lot	3.3, 4.3.3	(M101)
Plug Assembly	Insulation resistance	3.4, 4.7.1	(M102)
Bridgewire	Soldering	3.5	(M103)
Bridgewire Assembly	Continuity	3.6, 4.7.2	(M104)
Flash Charge	Compounding and loading	3.7	(M105)
Ignition Charge	Compounding and loading	3.8, 3.8.1	(M106)
Booster and Main Charges	Acceptance and loading	3.9.1	(M107)
Booster and Main Charges	Weighing	3.9.2	(M108)
Detonator	Visual inspection	3.10.1, 4.7.4	(M109)
Detonator	Bridge circuit resistance	3.10.2, 4.7.3	(M110)
Detonator	X-ray	3.10.3, 4.7.5	(M111)
Detonator	Ballistic requirements	3.11, 4.7.6	(M112)
Detonator	Packaging, packing and container markings	4.6.1	(M113)

4.6.1 Packaging, packing and container markings. The packaging and packing of detonators and the container markings shall be in accordance with Section 5. Quality conformance inspection shall be conducted in accordance with the inspection criteria on the applicable drawings referenced in Section 5.

4.7 Tests.

4.7.1 Insulation resistance. Prior to assembly of the bridgewire across the ground ends of the lead wires of the Plug and Bridgewire Assembly, Drawing 2299543, the insulation resistance between the lead wires shall be checked with an applied voltage of 500 ± 25 volts DC. Units which fail to meet the requirements of 3.4 shall be rejected.

4.7.2 Bridgewire assembly, continuity. The assembly shall be thoroughly cleaned to remove flux and contamination. After cleaning, the bridgewire shall be flashed by application of an electric current to assure continuity of the bridge. The current shall be regulated to produce a visible red glow of the bridge. An electromotive force of approximately

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1.5 volts AC or DC may be used with a suitable resistor to adapt the source characteristics to the bridge resistance. Flashing is a necessary manufacturing operation and shall be conducted on each bridgewire assembly. Assemblies which fail to glow shall be rejected. The faulty assemblies may be reworked by replacement of the bridges.

4.7.3 Bridge circuit resistance. Conformance with the bridge circuit resistance of 3.10.2 shall be ascertained on all completely assembled detonators by means of a test circuit which limits the bridge current to 25 milliamperes, maximum. A safety chamber with an interlock switch shall be used to protect the operator during the application of the current. A suitable instrument which is known to be accurate within one percent in the range of resistance specified shall be used to determine whether or not the requirement is being met. Units which fail to meet the requirements of 3.10.2 shall be rejected.

4.7.4 Visual inspection. Examine all completely assembled detonators and reject those having any of the defects noted in 3.10.1.

4.7.5 X-ray examination. Examine all detonators by X-ray and any observable imperfections in assembly or evidence of missing charges shall be cause for rejection.

4.7.6 Ballistic test.

4.7.6.1 The number of detonators selected from each production lot for the ballistic test shall be in accordance with the requirements of Tables A-2 and B-1 of MIL-STD-414, Inspection Level IV, for an AQL of 0.065. The detonators shall be conditioned at $70^{\circ} \pm 5^{\circ}$F for a minimum of six hours and fired in accordance with the detonator output test as specified in Test Number 301 of MIL-STD-331. The detonator confining sleeve material, dimensions and fit with the detonators shall be as specified in MIL-STD-331 except that only brass material shall be used. Instrumentation shall be provided to measure the time from application of firing current to the instant of case opening. The depth of indentation produced in the steel block by the detonator output shall be measured as specified in MIL-STD-331. The firing current shall be 5.0 ± 0.3 amperes DC. In addition to the sample of detonators for the above test, 10 additional detonators shall be furnished from each production lot for investigative purposes.

4.7.6.2 Criteria of acceptability.

4.7.6.2.1 Output. The lower (L) limit of depth of indentation produced by the detonator output shall be as specified in 3.11.1. With respect to output, the lot shall be acceptable if the quantity (X-L)/s is equal to or greater than the appropriate value of k, at an AQL of 0.065, when computed in accordance with MIL-STD-414, Part I, Table B-1.

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4.7.6.2.2 Misfire. The failure of any detonator to fire shall result in rejection of the lot represented, unless such failure is plainly attributable to faulty test equipment or procedure.

4.7.6.2.3 Hangfire. One or more hangfires shall result in rejection of the lot represented. A hangfire is defined as a firing in which the elapsed time between application of current and the instant of case opening exceeds 0.050 second.

4.7.7 Retest. There shall be no retests.

4.7.7.1 Test failure. If test failure is attributable to an assignable cause, excluding the test detonators, the original test results should be discarded and that portion of the test reconducted.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. For definitions of packaging levels, see 6.4.

5.1.1 Level A. Level A packaging shall be used for packaging of all production lots for service use. Four detonators shall be packaged in an inner container in accordance with LD 537918.

5.1.2 Level B. Not applicable.

5.1.3 Level C. When directed by the procuring activity, those detonators which are intended for expenditure in preproduction sample testing and for production lot acceptance tests shall be packaged in accordance with Method III of MIL-P-116 and with the Code of Federal Regulations 49 CFR 71-78.

5.2 Packing. For definitions of packing levels, see 6.4.

5.2.1 Level A. Level A packing shall be used for packing of all production lots for service use. Six inner containers as described in 5.1.1 shall be packed in accordance with LD 537921.

5.2.2 Level B. Not applicable.

5.2.3 Level C. When directed by the procuring activity, those detonators which are intended for expenditure in preproduction sample testing and for production lot acceptance tests and are packaged in accordance with 5.1.3 shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall be in accordance with the Code of Federal Regulations 49 CFR 71-78.

5.3 Markings.

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5.3.1 Special markings. Marking of exterior containers shall be in accordance with the Code of Federal Regulations 49 CFR 71-78.

5.3.2 Normal markings. Unless otherwise specified in the contract or order, the marking information on unit packages and shipping containers shall be as specified on the drawings of LD 537918 and LD 537921, respectively. The specified marking information shall be applied to the containers in accordance with the applicable provisions of MIL-STD-129.

6. NOTES

6.1 Intended use. Primary application of the MARK 83 MOD 0 electric detonator is to burst the plastic nose cone components of the MARK 12 MOD 0 aircraft chemical tank.

6.2 Ordering information. Procurement documents should specify the following:

a. title, number and date of this specification;

b. whether preproduction inspection is required and if so, specify the test activity;

c. production lot size and test activity (see 4.3.3);

d. items of data required for each preproduction and production lot (see 6.3).

6.3 Data. For the information of Contractors and Contracting Officers, any of the data specified in (a) subparagraphs below, (b) applicable documents listed in Section 2 of this specification or (c) referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order. The data to be furnished shall be listed on DD Form 1423 (Contractor Data Requirements List), which shall be attached to and made a part of the contract or order. NAVWEPS Form 4200/15 (Drawings, Lists, and Specifications Required) shall be attached where applicable.

6.3.1 Test report. Results of ballistic tests shall be forwarded to the Naval Air Systems Command and the procuring activity.

6.3.2 Data cards. The contract or order should state the format, quantity and destination of data cards.

6.4 Definitions. The definitions of Levels A and C packaging and packing given below were extracted from BUSANDA Instruction 4030.17.

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6.4.1 Level A packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during handling, shipment, indeterminate storage and world-wide redistribution.

6.4.1.1 Level C packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity, for immediate use. This level may conform to supplier's commerical practice when such meets the requirements of this level.

6.4.2 Level A packing. The degree of packing which will afford adequate protection during shipment, handling, indeterminate storage and world-wide redistribution.

6.4.2.1 Level C packing. The degree of packing which will afford protection against damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level in general will conform to applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirements of this level.

6.5 Explosives safety precautions.

6.5.1 Applicability. The loading, assembly and handling of the explosives, subassemblies and the finished items covered by this specification involve hazardous operations and therefore require suitable explosive safety precautions. At plants of the Naval Air Systems Command, Department of the Navy, and other Government plants where the Government is responsible for safety, the precautions listed below are mandatory. These same minimum provisions are recommended for observance by a contractor or manufacturer that explosively loads the item covered in this specification in order to fulfill his own safety responsibility. Other warnings and precautions, pertaining to the operational effectiveness or safety of the specified item during its use (for example, during its handling, launching or firing), are included in the detailed technical requirements of the specification. Use of this specification shall not be construed to replace other safety regulations of governmental agencies or to relieve the contractor or manufacturer of responsibility for the safety of his operations.

6.5.2 Personnel and equipment. Proper care must be exercised at all times to protect personnel from accidents, fires or explosions and to limit damage to equipment and loading areas. In connection with the above, the precautionary measures in the following paragraphs should be observed. (See also paragraph 6.5.4, Static and electrical and other applicable parts of these safety precautions.)

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6.5.2.1 Explosives operations shall be kept under strict supervision. Periodic inspections shall be employed to assure compliance with established safety regulations.

6.5.2.2 Safe equipment and methods should be utilized for handling and transporting explosives and loaded parts.

6.5.2.3 Where required, remote control-barricaded-handling equipment shall be used for explosives operations such as mixing, pouring, weighing, charging, sifting, drying, pressing, casting, crimping, etc.

6.5.2.4 Employ properly proportioned and properly located protective barricades, screens or shields at all required points.

6.5.2.5 Maintain equipment such as presses, dies, crimpers, blenders, mixers, sifters, conveyors, and the like in safe operating condition.

6.5.2.6 Enforce, where necessary, the wearing of suitable safety footwear, gloves, goggles, respirators, and impregnated garments to protect personnel against burns, poisoning and associated industrial hazards.

6.5.2.7 During static tests, propulsion units such as rocket motors shall be restrained against full thrust to prevent flight if ignited, for example, during test of igniter circuit. Also, means shall be provided to harmlessly vent flames and exhaust gases.

6.5.3 Operating procedures. All loading operations shall be conducted in full accordance with a formalized operating procedure which shall include the following:

6.5.3.1 Keep only minimum, limited quantities of explosives and completed or partially loaded parts present at each stage of operation.

6.5.3.2 Allow no fires, exposed electrical or other sparking equipment, and little or no flammable material to be present in loading, handling and storage spaces. Enforce proper "Match" and "No Smoking" rules where necessary.

6.5.3.3 Appropriate safety instructions for specific operations shall be posted and personnel instructed as to the hazards present and precautions involved.

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6.5.3.4 The number of working personnel and transients permitted in each work space shall be established and posted in such areas. The numbers shall be the minimum number required to accomplish necessary operations.

6.5.3.5 All loading operations should be conducted in a neat and orderly manner.

6.5.3.6 Enforce good housekeeping and maintain effective policing, inspection and supervisory methods throughout the loading area and surroundings.

6.5.3.7 Employ effective cleaning methods periodically to minimize the accumulation of explosives or explosive dust and other contamination upon, and assure its removal from, floors, walls, ceiling, ledges, tables, benches, piping, and equipment or the items loaded. Clean up any spilled material immediately.

6.5.3.8 The exposure of explosive materials and related parts should be so controlled as to minimize the absorption of moisture from the atmosphere or other sources during loading and handling operations.

6.5.3.9 Personnel handling detonators, primers, delay elements, lead-ins, boosters, and related parts which affect functioning should, insofar as practicable, avoid using bare fingers or improper equipment in order to prevent damage, corrosion, or deterioration from perspiration or other contaminating deposits.

6.5.3.10 Keep explosives and explosive parts in approved covered receptacles with covers in place when material is not being taken out or put into the receptacles. Where necessary, receptacles should be conductive to ground electrostatic charges. (See paragraph 6.5.4.)

6.5.4 Static and electrical.

6.5.4.1 Protect operations from electrostatic charges by effectively grounding all machinery, equipment, and fixtures; and, where necessary, employ suitable grounded conductive coverings for floors, work benches and tables, and workers' conductive shoes. Workers' clothing of a type to minimize the accumulation of static charges shall be employed. Fabrics, such as silk and nylon, which promote static generation shall be avoided. Additional grounding devices such as grounded bracelets for workers should be employed where operations are conducted with items which are unusually

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sensitive to initiation by static electricity. Such items include initiating explosives, tracer mixtures, and low energy type of type electric primers, detonators and squibs. The latter types of items shall have the free ends of lead wires bared and twisted together, and be packed in relatively small groups wrapped in bare non-insulated aluminum foil or other uncoated metal foil. During assembly and processing operations, such sensitive electric items should be short circuited by clips or other devices until installed with safety shunt in the final device. Additional precautions for these items should include mechanical shielding to contain or deflect fragments and blast; also electrical shielding from induced electric currents originated by sources such as lightning, static, electromagnetic radiations, etc. Where necessary for safety, humidity of work rooms should be appropriately increased, as required to lessen electrostatic effects, but without inducing excessive moisture absorption by any of the components of the item being processed.

6.5.4.2 Protect all explosive operations from effects of electric current originating in equipment (such as soldering irons or guns, heaters, switches, wiring, motors, lights, test instruments, etc.) by suitable insulation, grounding, separation or shielding. Such electric sources may initiate explosives by heat, sparks, arcs, or by completing an electric circuit through an electric primer, detonator or squib. For example, circuits may be inadvertently completed from a defective electric soldering iron through a grounded contact. All electric type of primers, detonators or squibs provided with wire leads shall have the free ends of the wires bared and twisted together to short circuit each unit, except when in process of assembly into a finished item. Removable short circuiting clips or other devices should be employed during manufacturing operations involving items such as electric primers, detonators or squibs.

6.5.4.3 Where necessary, protect personnel and operations from electromagnetic energy. In certain work with very sensitive explosives or items, a shielded environment free from significant electromagnetic energy should be provided.

6.5.5 Operating facility and storage regulations.

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6.5.5.1 All explosives including partially loaded items should be stored in suitable storage magazines located in accordance with the minimum distances prescribed in the American Table of Distances (ATD). While the explosive or loaded items are in process, stored in loading rooms, or in ready or service magazines, they should be located in accordance with Intraplant Distances set forth in the ATD. For Navy managed or contractor managed-Navy owned-explosives loading plants, the Armed Services Explosives Safety Board standards approved by the Department of Defense, the Secretary of the Navy and the Naval Air Systems Command covering quantity-distance regulations for explosives apply.

Project No. 1377-N465

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-D-81549A(AS) DETONATOR, ELECTRIC, MARK 83 MOD 0

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426

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Naval Air Systems Command
Washington, D.C. 20360

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